MAHATMA GANDHI UNIVERSITY, KOTTAYAM

25900353: Analog Electronics (MG3DSCECT200 / MG3DSCECC200)

Exam: Regular Examination, October 2025 | Semester: III – B.Sc. (Honours)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Scheme of Valuation

Part A – Multiple Choice Questions

(10 × 1 = 10 Marks)

Each question carries **1 mark**. Award full marks for the correct option.

1. In a voltage divider network, collector current Ic is mainly determined by Re.
   (c) Re — 1 mark
2. The Barkhausen criterion requires the feedback signal to be in phase with the input.
   (a) Phase with the input — 1 mark
3. In an RC differentiator, the output voltage is proportional to the derivative of the input voltage.
   (c) — 1 mark
4. In a BJT, the emitter region is heavily doped.
   (b) Emitter — 1 mark
5. The minimum closed-loop gain of a non-inverting amplifier is 1.
   (c) 1 — 1 mark
6. The input offset voltage is the voltage needed between inputs to make output zero.
   (b) — 1 mark
7. In a 741 op-amp, PIN 7 is used for positive power supply (V+).
   (c) — 1 mark
8. The loop gain requirement for sustaining oscillations in an RC phase-shift oscillator is $A \geq 29$.
   (c) — 1 mark
9. In a non-inverting Schmitt trigger, the input signal is applied to the non-inverting terminal.
   (b) — 1 mark
10. In an op-amp comparator, no feedback is used.
    (c) — 1 mark

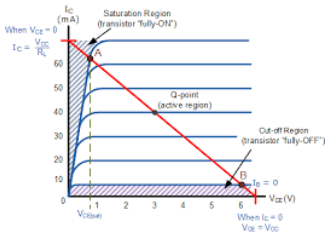
Total: 10 Marks

Part B – Short Answer Questions

(Answer any four. Each question carries 5 marks 5 × 4 = 20 Marks)

Q11. *Why RC coupling is preferred in multistage amplifier design?*
RC coupling is widely used because it offers excellent frequency response, a simple construction, low cost, and a compact size. It passes AC signals while blocking DC, ensuring each stage is properly biased without affecting the next. It also maintains good signal fidelity and has minimal distortion.
Marks distribution: Concept (2), Advantages (3) = 5 marks

Q12. *Briefly explain the saturation mode of operation of a BJT.*
In saturation mode, both the base–emitter and base–collector junctions are forward biased. The transistor conducts heavily, and collector current is maximum and independent of base current. Collector-emitter voltage (Vce) drops to a low value (~0.2V). The transistor behaves like a closed switch.



Marks distribution: Definition (2), Junction condition (1), Characteristics (2) = 5 marks

Q13. *Compare inverting and non-inverting amplifiers in terms of gain and phase shift.*
An inverting amplifier produces an output that is 180° out of phase with the input and has a gain of $A_v = -\frac{R_f}{R_{in}}$.
A non-inverting amplifier produces an output in phase with the input and has a gain of

$$A_v = 1 + \frac{R_f}{R_{in}}.$$

The non-inverting configuration offers higher input impedance and no phase reversal, whereas the inverting configuration offers controlled gain with phase inversion.
Marks distribution: Explanation (2), Equations (2), Phase relation (1) = 5 marks

Q14. *Draw the block diagram of an Op-Amp.*
An op-amp consists of several stages:

- Input differential amplifier (for high input impedance and differential gain)

- Intermediate gain stage (amplifies further)
- Level shifter (centers output voltage)
- Output stage (provides low output impedance)

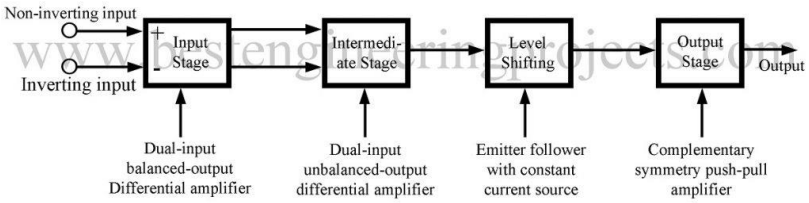


Figure 1: Block Diagram of a Typical Op-amp

Marks distribution: Diagram (3), Explanation (2) = 5 marks

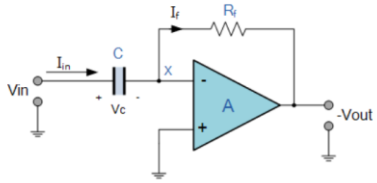
Q15. *Why is the open-loop op-amp gain considered impractically high for analog signal amplification?*

The open-loop voltage gain of an op-amp (typically 10^5–10^6) is so high that even a small input differential voltage drives the output to saturation (±V_sat). Hence, direct analog amplification is not practical. To ensure linear operation and controlled gain, feedback (usually negative) is introduced.

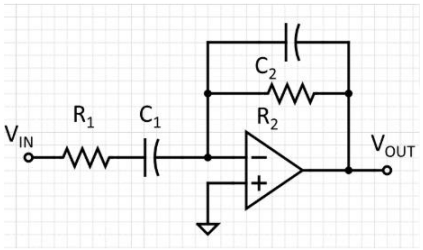
Marks distribution: Concept (2), Reason (2), Solution (1) = **5 marks**

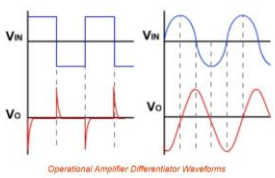
Q16. *Explain how a differentiator circuit can be constructed using an op-amp. Give its output waveform.*

An op-amp differentiator is a circuit that produces an output voltage proportional to the rate of change of its input voltage, essentially performing a mathematical differentiation. It is an inverting amplifier with a capacitor at the input and a feedback resistor, which results in the formula $V_{out} = -RC\frac{dV_{in}}{dt}$.



Ideal Differentiator





Marks distribution: Circuit diagram (2), Working (2), Waveform (1) = **5 marks**

Total: 20 Marks

Part C – Essay Questions

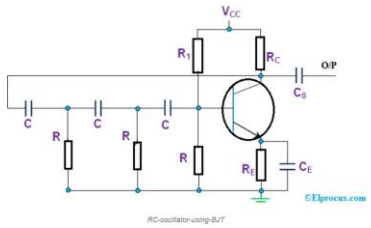
(Answer any two. Each carries 10 marks 10 × 2 = 20 Marks)

Q17. *Explain the working principle of an RC phase shift oscillator using a BJT.*
An RC phase shift oscillator consists of a transistor amplifier and a feedback network of three RC sections, each providing 60° phase shift, totaling 180°. The amplifier contributes an additional 180°, ensuring the total phase shift of 360° (Barkhausen criterion satisfied).

Condition for oscillation $A\beta \geq 1$, where $A\beta = 29$ for the RC network.

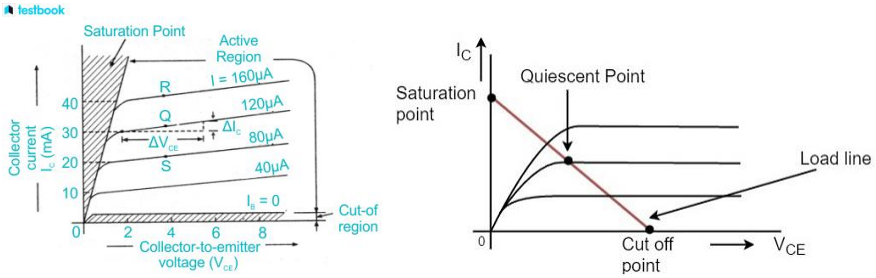
Frequency of oscillation:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$



Marks distribution: Circuit (3), Working (3), Condition and formula (2), Diagram clarity & conclusion (2) = **10 marks**

Q18. *What is the operating point of a BJT? Explain why it is important.*
The operating point (Q-point) represents the steady-state values of collector current (Ic) and collector-emitter voltage (Vce) when no input signal is applied. It defines the transistor's DC bias condition.
A proper Q-point ensures the transistor operates in the active region for faithful amplification. If not chosen properly, the output may distort due to cut-off or saturation. Graphical method using the DC load line can be used to locate the Q-point.



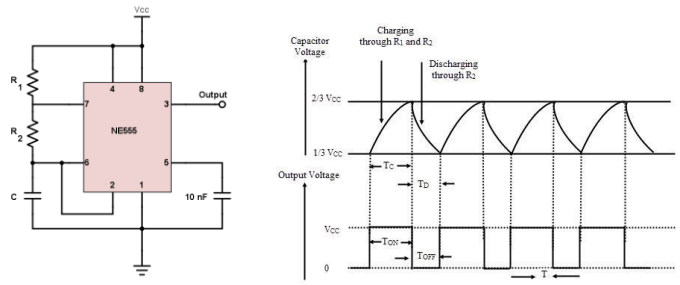
Marks distribution: Definition (2), Importance (3), Explanation with load line (3), Diagram & clarity (2) = **10 marks**

Q19. *Compare the transition from SSI to VLSI in Integrated Circuits.*

- SSI (Small Scale Integration): up to 100 components per chip; used in basic logic gates.
- MSI (Medium Scale Integration): 100–1,000 components; used in counters and registers.
- LSI (Large Scale Integration): 1,000–10,000 components; used in simple processors.
- VLSI (Very Large Scale Integration): >10,000 components; used in microprocessors, memory chips.
  The evolution from SSI to VLSI led to smaller size, lower power consumption, higher speed, and improved reliability. Moore's Law describes the exponential growth of integration levels.
  Marks distribution: Explanation of stages (6), Comparison & examples (2), Evolution impact (2) = **10 marks**

Q20. *Describe the astable mode operation of the 555 timer IC with circuit example. Write the equations for frequency and duty cycle.*
In astable mode, the 555 timer operates as a free-running multivibrator. Two resistors (RA, RB) and a capacitor (C) determine the charging and discharging times.
The output continuously switches between high and low without any external trigger.

- Frequency: $f = \frac{1.44}{(R_A+2R_B)C}$
- Duty cycle: $D = \frac{R_A+R_B}{R_A+2R_B}$



Marks distribution: Circuit (3), Working (3), Formulae (2), Waveform & explanation (2) = **10 marks**